

MANOHAR NAGA

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Education

IIIT Hyderabad

Nov 2021 – May 2025

B.Tech in Computer Science Engineering, CGPA: 8.54/10.0

Experience

Microsoft

Jul 2025 – Present

Software Engineer

Hyderabad, India

- Optimized large-scale **PySpark** data pipelines processing **150K+ recommendations**, reducing execution time from **20 hours to 20–25 minutes (~98% faster)** through bulk joins, cached Spark execution, and vectorized writes.
- Designed and shipped AI-powered customer summary services for Azure customers, reducing summary generation latency from **5–10 minutes to < 1 second** using scheduled pre-computation, Cosmos DB caching, and distributed processing.
- Improved backend API performance by reducing **P95 latency by 21.3%** and **P99 latency by 24.6%** across 44 production APIs through profiling, query optimization, and infrastructure improvements.

Game Theory

May 2024 – Jul 2024

Software Engineer Intern

Hyderabad, Telangana

- Built **4+ annotation tools** from ground up for object and human tracking, supporting both video and image inputs.
- Achieved **25x** reduction in frame load latency (from **5s to 0.2s**), a significant performance boost over previous tools.
- Optimized UI for annotation tools, improving workflow efficiency and reducing annotation time by **4x** (from **16 hours to 4 hours** per video(7500+ frames)).

Crop Darpan

Jan 2024 – May 2024

BTech Project - Data Sciences and Analytics Center

Hyderabad, Telangana

- Designed and developed a Firebase-based **user log** storage system, integrated it into the existing app, addressing **Expo SDK 48** limitations through dependency resolution and streamlined **APK** generation, added comprehensive documentation for maintenance.

Teaching Assistant

Jan 2024 – Mar 2024

CS8.301 Introduction to Information Security

- Prepared problem sets, conducted exams, tutorials and office hours for the overall comprehensive learning of the students.

Shadow Media Productions

Feb 2023 – May 2023

Full Stack Developer Intern

Hyderabad, Telangana

- Developed **movies page(both UI and backend)**, which supports all CRUD operations for a movie.
- Worked with **google firebase** to build high quality movie posters page(**15MB+**) and authentication page which supports SignIn/SignUp with Gmail/PhoneNo.

Projects

DataSphere | BUC, AOI, Apriori

[Code](#)

- Developed an OLAP project that extracted interesting datacubes using the BUC algorithm with optimizations (external sorting, hash partitioning, iceberg cubes) and derived characteristic rules through attribute-oriented induction. Developed a Movie recommender system using Apriori Algorithm.

MemeMaker | React-Native, APIs

[Code](#)

- Designed and developed a MemeMaker mobile app, featuring a canvas for users to add/rotate/resize text, integrated Stable Diffusion model API to generate meme images from text input.

Greddit | MERN Stack, Firebase, Docker

[Code](#)

- Developed a full-stack social media MERN app like Reddit and dockerised it for easy deployment.
- Key features include - authentication(Firebase), authorization(JWT), profile and subreddit pages, post interactions (like,comment), large profile picture uploads (15MB+), user joining requests, admin controls, banned keywords, fuzzy search and more.

Technical Skills

Languages: C/C++, C#, Python3, HTML/CSS, JavaScript, SQL, KQL, Bash

Technologies/Frameworks: Azure CosmosDB, Azure DevOps, .NET8, React, Node.js, Next.js, MongoDB, Git/Github

Fundamentals/Tools: DBMS, Operating Systems, TCP/IP, OOPs, GNU/Linux, Windows

Achievements

- 2+ years** experience in the Industry.
- Progressed to *Round 2* in **MetaHackerCup2023** - Global Rank(2934).
- Got Placed in **Dean's List**, top **5%** of our batch for Academic Excellence in Monsoon 2024.